

Supplementary Online Content

Kong X, Liu Z, Cheng R, et al. Variation in breast cancer subtype incidence and distribution by race/ethnicity in the United States from 2010 to 2015. *JAMA Netw Open*. 2020;3(10): e2020303. doi:10.1001/jamanetworkopen.2020.20303

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This supplementary material has been provided by the authors to give readers additional information about their work.

eTable 1. Number and Percentage of Patients with Each Breast Cancer Molecular Subtype by Race/Ethnic Group and Case-Case ORs for Each Subtype Compared With HR+/HER2-

Molecular subtype (BreastType)	Non-Hispanic White No. (%)	Black			Asian/Pacific Islander			Hispanic White			American Indian/Alaskan Native		
		No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI
HR+/HER2-	122295 (75.3)	16372 (60.8)	Ref.	-	15067 (71.5)	Ref.	-	18872 (68.8)	Ref.	-	969 (69.1)	Ref.	-
HR+/HER2+	16902 (10.4)	3213 (11.9)	1.28	1.23 to 1.34	2660 (12.6)	1.19	1.14 to 1.25	3482 (12.7)	1.2	1.15 to 1.25	188 (13.4)	1.36	1.16 to 1.59
HR-/HER2+	6739 (4.2)	1640 (6.1)	1.64	1.55 to 1.74	1436 (6.8)	1.66	1.56 to 1.76	1585 (5.8)	1.41	1.33 to 1.5	80 (5.7)	1.47	1.17 to 1.85
Triple Negative	16423 (10.1)	5713 (21.2)	2.4	2.31 to 2.48	1923 (9.1)	0.91	0.87 to 0.96	3486 (12.7)	1.28	1.23 to 1.34	166 (11.8)	1.26	1.07 to 1.49

Abbreviations: NHW, Non-Hispanic White.

*Adjusted for age, SEER registry region, and year of diagnosis. Case-case OR = (%subtype/%HR+/HER2-)subpop/(%subtype/%HR+/HER2-)NHW.

eTable 2. Number and Percentage of Patients with Each Breast Cancer Histological Grade by Race/Ethnic Group and Case-Case ORs for Each Grade Compared With Grade I

Histological Grade (Grade)	Non-Hispanic White No. (%)	Black			Asian/Pacific Islander			Hispanic White			American Indian/Alaskan Native		
		No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI
Well differentiated; Grade I	39267 (24.2)	3736 (13.9)	Ref.	-	4102 (19.5)	Ref.	-	4979 (18.2)	Ref.	-	296 (21.1)	Ref.	-
Moderately differentiated; Grade II	70336 (43.3)	9646 (35.8)	1.36	1.31 to 1.42	9091 (43.1)	1.26	1.21 to 1.31	11222 (40.9)	1.25	1.21 to 1.3	558 (39.8)	1.09	0.95 to 1.26
Poorly differentiated; Grade III	46819 (28.8)	12177 (45.2)	2.4	2.3 to 2.49	7078 (33.6)	1.4	1.34 to 1.46	10138 (37)	1.56	1.51 to 1.62	488 (34.8)	1.4	1.21 to 1.62
Undifferentiated; anaplastic; Grade IV	386 (0.2)	70 (0.3)	2.14	1.65 to 2.78	70 (0.3)	1.27	0.98 to 1.65	117 (0.4)	1.73	1.4 to 2.15	<=5	NE	
Unknown	5551 (3.4)	1309 (4.9)	2.08	1.94 to 2.23	745 (3.5)	1.63	1.5 to 1.78	969 (3.5)	1.64	1.52 to 1.78	59 (4.2)	1.91	1.44 to 2.53

Abbreviations: NHW, Non-Hispanic White.

*Adjusted for age, SEER registry region, and year of diagnosis. Case-case OR = (%subtype/%Grade I)_{subpop}/(%subtype/%Grade I)_{NHW}.

eTable 3. Number and Percentage of Patients with Each Breast Cancer Pathological Pattern by Race/Ethnic Group and Case-Case ORs for Each Pattern Compared With Infiltrating Duct Carcinoma

Pathological Pattern (HistBehav)	Non-Hispanic White No. (%)	Black			Asian/Pacific Islander			Hispanic White			American Indian/Alaskan Native		
		No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI
8500/3: Infiltrating duct carcinoma	121878 (75.1)	21040 (78.1)	Ref.	-	16840 (79.9)	Ref.	-	21111 (77)	Ref.	-	1119 (79.8)	Ref.	-
8520/3: Lobular carcinoma	15718 (9.7)	1933 (7.2)	0.78	0.74 to 0.82	1202 (5.7)	0.58	0.55 to 0.62	1985 (7.2)	0.8	0.76 to 0.84	101 (7.2)	0.72	0.59 to 0.87
8522/3: Infiltrating duct and lobular carcinoma	9387 (5.8)	992 (3.7)	0.66	0.61 to 0.7	964 (4.6)	0.72	0.67 to 0.77	1709 (6.2)	1.04	0.99 to 1.1	83 (5.9)	0.95	0.77 to 1.18
8201/3: Cribriform carcinoma	282 (0.2)	40 (0.1)	0.86	0.62 to 1.19	43 (0.2)	1.28	0.91 to 1.8	48 (0.2)	1.14	0.83 to 1.58	<=5	NE	
8211/3: Tubular adenocarcinoma	997 (0.6)	81 (0.3)	0.42	0.34 to 0.54	55 (0.3)	0.45	0.34 to 0.59	88 (0.3)	0.55	0.44 to 0.69	<=5	NE	
8480/3: Mucinous adenocarcinoma	2911 (1.8)	554 (2.1)	1.23	1.12 to 1.35	503 (2.4)	1.5	1.36 to 1.66	471 (1.7)	1.17	1.06 to 1.29	15 (1.1)	0.69	0.43 to 1.13
8523/3: Infiltrating duct mixed with other types of carcinoma	5265 (3.2)	903 (3.4)	1	0.93 to 1.08	709 (3.4)	1.05	0.97 to 1.14	922 (3.4)	1.1	1.02 to 1.18	31 (2.2)	0.7	0.5 to 0.99
8507/3: Ductal carcinoma, micropapillary	607 (0.4)	122 (0.5)	1.26	1.03 to 1.54	101 (0.5)	1.17	0.95 to 1.45	118 (0.4)	1.02	0.83 to 1.26	<=5	NE	

Other	5314 (3.3)	1273 (4.7)	1.38	1.29 to 1.47	669 (3.2)	1.05	0.96 to 1.14	973 (3.5)	1.2	1.12 to 1.29	48 (3.4)	1.05	0.79 to 1.4
Abbreviations: NHW, Non-Hispanic White. *Adjusted for age, SEER registry region, and year of diagnosis. Case-case OR = (%subtype/%Infiltrating Duct Carcinoma)subpop/(%subtype/%Infiltrating Duct Carcinoma)NHW.													

eTable 4. Number and Percentage of Patients with Each T Stage of Breast Cancer by Race/Ethnic Group and Case-Case ORs for Each T Stage Compared With T1

Tumor Size (T Stage)	Non-Hispanic White No. (%)	Black			Asian/Pacific Islander			Hispanic White			American Indian/Alaskan Native		
		No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI
T1	99334 (61.2)	13099 (48.7)	Ref.	-	11673 (55.4)	Ref.	-	14166 (51.7)	Ref.	-	782 (55.7)	Ref.	-
T2	47397 (29.2)	9655 (35.9)	1.51	1.47 to 1.56	7273 (34.5)	1.2	1.16 to 1.24	9850 (36)	1.31	1.27 to 1.35	457 (32.6)	1.16	1.03 to 1.3
T3	9466 (5.8)	2368 (8.8)	1.84	1.75 to 1.94	1321 (6.3)	0.99	0.93 to 1.05	2156 (7.9)	1.29	1.23 to 1.36	98 (7)	1.15	0.93 to 1.42
T4	5993 (3.7)	1763 (6.6)	2.18	2.06 to 2.31	800 (3.8)	1.18	1.09 to 1.28	1212 (4.4)	1.45	1.35 to 1.55	66 (4.7)	1.5	1.16 to 1.94

Abbreviations: NHW, Non-Hispanic White.

*Adjusted for age, SEER registry region, and year of diagnosis. Case-case OR = (%subtype/%T1)subpop/(%subtype/%T1)NHW.

eTable 5. Number and Percentage of Patients with Each AJCC TNM Stage Group of Breast Cancer by Race/Ethnic Group and Case-Case ORs for Each Group Compared With Stage I

AJCC Stage Group (Stage)	Non-Hispanic White No. (%)	Black			Asian/Pacific Islander			Hispanic White			American Indian/Alaskan Native		
		No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI
Stage I	86220 (53.1)	10700 (39.7)	Ref.	-	10128 (48)	Ref.	-	11582 (42.2)	Ref.	-	648 (46.2)	Ref.	-
Stage II	52479 (32.3)	10339 (38.4)	1.54	1.5 to 1.59	7804 (37)	1.14	1.1 to 1.18	10656 (38.9)	1.33	1.3 to 1.37	506 (36.1)	1.2	1.06 to 1.35
Stage III	16911 (10.4)	4084 (15.2)	1.84	1.77 to 1.92	2369 (11.2)	1.06	1.01 to 1.11	3970 (14.5)	1.51	1.44 to 1.57	182 (13)	1.33	1.12 to 1.57
Stage IV	6241 (3.8)	1725 (6.4)	2.1	1.99 to 2.23	731 (3.5)	0.97	0.89 to 1.05	1119 (4.1)	1.27	1.18 to 1.36	65 (4.6)	1.39	1.07 to 1.8
UNK Stage	508 (0.3)	90 (0.3)	1.53	1.21 to 1.92	54 (0.3)	1.15	0.86 to 1.54	98 (0.4)	1.84	1.46 to 2.31	<=5	NE	

Abbreviations: NHW, Non-Hispanic White.

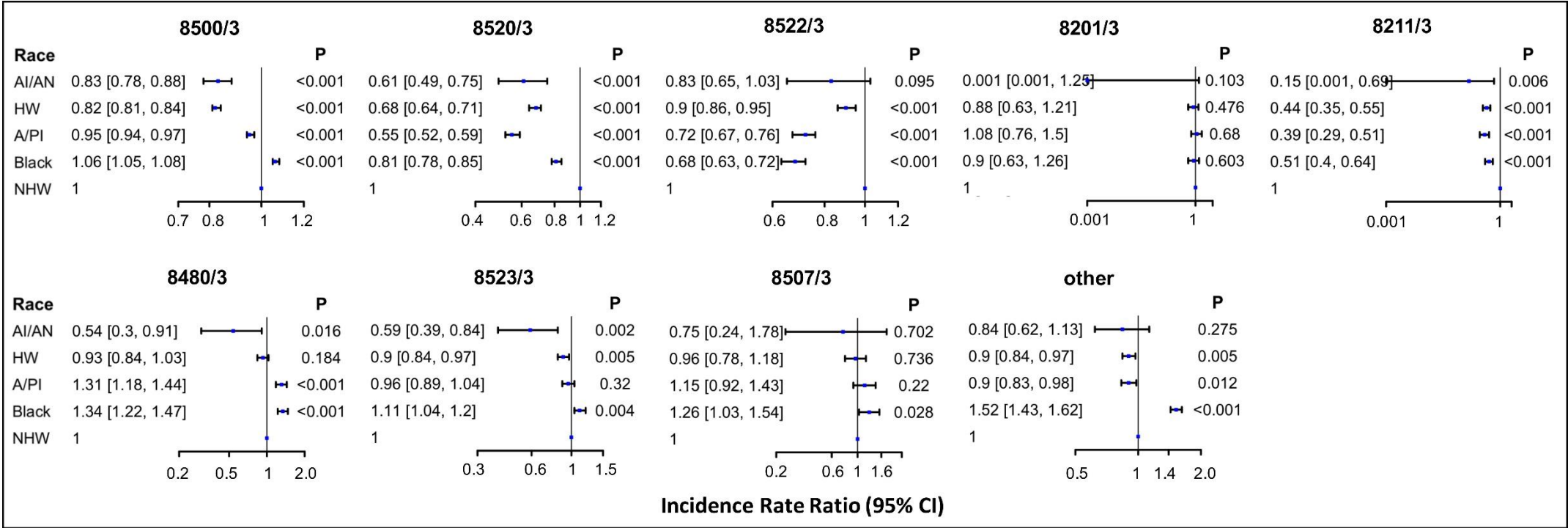
*Adjusted for age, SEER registry region, and year of diagnosis. Case-case OR = (%subtype/%Stage I)subpop/(%subtype/%Stage I)NHW.

eTable 6. Number and Percentage of Patients with Each Breast Cancer Tumor Site by Race/Ethnic Group and Case-Case ORs for Each Site Compared with Upper-Outer Quadrant of Breast

Tumor Site (PrimarySiteLabeled)	Non-Hispanic White No. (%)	Black			Asian/Pacific Islander			Hispanic White			American Indian/Alaskan Native		
		No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI	No. (%)	OR*	95% CI
Upper-outer quadrant	57308 (35.3)	9337 (34.7)	Ref.	-	6789 (32.2)	Ref.	-	9248 (33.7)	Ref.	-	487 (34.7)	Ref.	-
Upper-inner quadrant	19965 (12.3)	3325 (12.3)	1.05	1 to 1.1	3041 (14.4)	1.26	1.21 to 1.33	3429 (12.5)	1.05	1.01 to 1.1	212 (15.1)	1.22	1.04 to 1.44
Lower-outer quadrant	12452 (7.7)	1953 (7.2)	0.98	0.93 to 1.03	1516 (7.2)	1	0.94 to 1.06	2006 (7.3)	0.97	0.92 to 1.02	97 (6.9)	0.89	0.71 to 1.11
Lower-inner quadrant	9093 (5.6)	1711 (6.4)	1.17	1.11 to 1.24	1140 (5.4)	1.12	1.05 to 1.2	1418 (5.2)	1.03	0.97 to 1.09	59 (4.2)	0.8	0.61 to 1.05
Central portion + Nipple	8159 (5)	1133 (4.2)	0.87	0.81 to 0.93	1196 (5.7)	1.42	1.33 to 1.52	1310 (4.8)	1.15	1.08 to 1.23	80 (5.7)	1.33	1.05 to 1.69
Overlapping lesion of breast	37215 (22.9)	5881 (21.8)	0.99	0.96 to 1.03	5104 (24.2)	1.15	1.11 to 1.2	6590 (24)	1.1	1.06 to 1.14	294 (21)	0.92	0.8 to 1.07
Axillary tail of breast	771 (0.5)	182 (0.7)	1.46	1.24 to 1.73	81 (0.4)	0.79	0.62 to 1	130 (0.5)	0.92	0.76 to 1.12	<=5	NE	

Breast, NOS	17396 (10.7)	3416 (12.7)	1.2	1.15 to 1.25	2219 (10.5)	1.04	0.99 to 1.1	3294 (12)	1.13	1.08 to 1.18	172 (12.3)	1.14	0.96 to 1.36
Abbreviations: NHW, Non-Hispanic White.													
*Adjusted for age, SEER registry region, and year of diagnosis. Case-case OR = (%subtype/%Upper-outer quadrant)subpop/(%subtype/%Upper-outer quadrant)NHW.													

eFigure 1. IRR of Breast Cancer Pathological Patterns by Race/Ethnicity Compared With NHWs



eFigure 2. IRR of Breast Cancer Tumor Locations by Race/Ethnicity Compared With NHWs

